

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AI)

DEEP LEARNING LAB RECORD

LAB - 03

Student Details

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Course Name: Deep Learning

Course Code: 22AIE304

Experiment Title

Advanced Hyperparameter Optimization for a Multilayer Perceptron using Grid Search

Aim

To optimize the performance of a Multilayer Perceptron (MLP) classifier by tuning its hyperparameters using GridSearchCV.

Objective

- Implement an MLP classifier.
 - Perform hyperparameter tuning using GridSearchCV.
 - Identify the optimal activation function, hidden layer size, and learning rate.
 - Evaluate the optimized model.
-

Dataset

Dataset Name: Iris Dataset

Number of Samples: 150

Number of Features: 4

Target Classes:

- Iris Setosa
 - Iris Versicolor
 - Iris Virginica
-

Software Requirements

- Python 3.x
 - NumPy
 - Pandas
 - Scikit-Learn
 - Matplotlib
 - Jupyter Notebook
-

Required Libraries

- numpy
- pandas
- matplotlib
- sklearn

```
In [31]: !pip install numpy pandas matplotlib scikit-learn
```

Requirement already satisfied: numpy in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (2.2.6)
Requirement already satisfied: pandas in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (2.3.3)
Requirement already satisfied: matplotlib in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (3.10.9)
Requirement already satisfied: scikit-learn in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (1.7.2)
Requirement already satisfied: pytz>=2020.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from pandas) (2026.2)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\msi\appdata\roaming\python\python310\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: tzdata>=2022.7 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from pandas) (2026.2)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from matplotlib) (4.63.0)
Requirement already satisfied: pyparsing>=3 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from matplotlib) (1.5.0)
Requirement already satisfied: cycler>=0.10 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: pillow>=8 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from matplotlib) (12.2.0)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from matplotlib) (1.3.2)
Requirement already satisfied: packaging>=20.0 in c:\users\msi\appdata\roaming\python\python310\site-packages (from matplotlib) (25.0)
Requirement already satisfied: joblib>=1.2.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: scipy>=1.8.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from scikit-learn) (1.15.3)
Requirement already satisfied: six>=1.5 in c:\users\msi\appdata\roaming\python\python310\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)

[notice] A new release of pip is available: 23.0.1 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip

```
In [32]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.model_selection import GridSearchCV

from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler

from sklearn.neural_network import MLPClassifier

from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
```

```
In [33]: from sklearn.datasets import load_iris
import pandas as pd

iris = load_iris(as_frame=True)
```

```
dataset = iris.frame

dataset.columns = [
    "SepalLength",
    "SepalWidth",
    "PetalLength",
    "PetalWidth",
    "Species"
]

dataset.head()
```

Out[33]:

	SepalLength	SepalWidth	PetalLength	PetalWidth	Species
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

In [34]:

```
X = dataset.iloc[:, :-1].values

y = dataset.iloc[:, -1].values

encoder = LabelEncoder()

y = encoder.fit_transform(y)

scaler = StandardScaler()

X = scaler.fit_transform(X)
```

In [35]:

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)
```

In [36]:

```
parameter_grid = {

    "hidden_layer_sizes":[(8,), (16,), (16,8)],

    "activation":["relu","tanh","logistic"],

    "learning_rate_init":[0.001,0.01,0.1],

    "max_iter":[500]

}
```

In [37]:

```
grid = GridSearchCV(
```

```

estimator=MLPClassifier(random_state=42),

param_grid=parameter_grid,

cv=5,

scoring="accuracy",

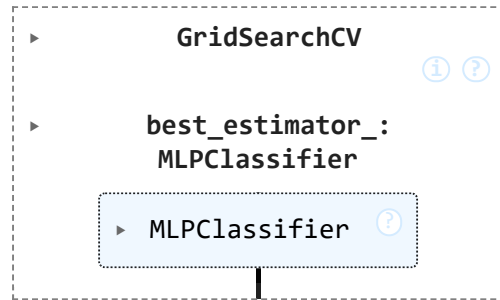
n_jobs=-1

)

grid.fit(X_train, y_train)

```

Out[37]:



```

In [38]: print("Best Parameters")

print(grid.best_params_)

print()

print("Best Accuracy")

print(grid.best_score_)

```

Best Parameters
{'activation': 'tanh', 'hidden_layer_sizes': (16,), 'learning_rate_init': 0.01, 'max_iter': 500}

Best Accuracy
0.9666666666666668

```

In [39]: best_model = grid.best_estimator_

predictions = best_model.predict(X_test)

```

```

In [40]: accuracy = accuracy_score(y_test, predictions)

print("Test Accuracy :", accuracy)

```

Test Accuracy : 0.9666666666666667

```

In [41]: print(confusion_matrix(y_test, predictions))

[[10  0  0]
 [ 0  9  1]
 [ 0  0 10]]

```

```

In [42]: print(classification_report(y_test, predictions))

```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	1.00	0.90	0.95	10
2	0.91	1.00	0.95	10
accuracy			0.97	30
macro avg	0.97	0.97	0.97	30
weighted avg	0.97	0.97	0.97	30

In [43]: `results = pd.DataFrame(grid.cv_results_)`

```
results[
    [
        "param_hidden_layer_sizes",
        "param_activation",
        "param_learning_rate_init",
        "mean_test_score"
    ]
].sort_values(
    by="mean_test_score",
    ascending=False
).head(10)
```

Out[43]:

	param_hidden_layer_sizes	param_activation	param_learning_rate_init	mean_test_score
--	--------------------------	------------------	--------------------------	-----------------

22	(16,)	logistic	0.010	0.96666
15	(16, 8)	tanh	0.001	0.96666
13	(16,)	tanh	0.010	0.96666
19	(8,)	logistic	0.010	0.96666
5	(16,)	relu	0.100	0.96666
16	(16, 8)	tanh	0.010	0.95833
14	(16,)	tanh	0.100	0.95833
10	(8,)	tanh	0.010	0.95833
12	(16,)	tanh	0.001	0.95833
6	(16, 8)	relu	0.001	0.95833



In [44]:

```
plt.figure(figsize=(10,5))

plt.plot(results["mean_test_score"])

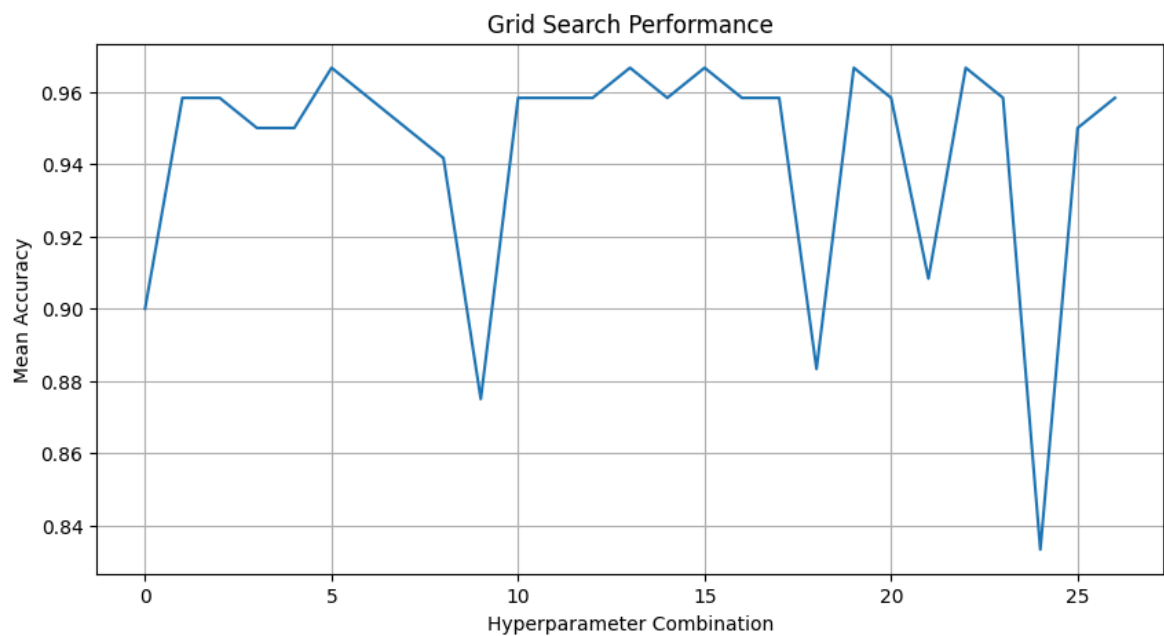
plt.xlabel("Hyperparameter Combination")

plt.ylabel("Mean Accuracy")

plt.title("Grid Search Performance")

plt.grid(True)

plt.show()
```



```
In [45]: results = pd.DataFrame(grid.cv_results_)

results = results[
    [
        "param_hidden_layer_sizes",
        "param_activation",
        "param_learning_rate_init",
        "mean_test_score",
        "rank_test_score"
    ]
]

results.sort_values("rank_test_score").head(5)
```

Out[45]:

	param_hidden_layer_sizes	param_activation	param_learning_rate_init	mean_test_score
13	(16,)	tanh	0.010	0.96666
15	(16, 8)	tanh	0.001	0.96666
22	(16,)	logistic	0.010	0.96666
19	(8,)	logistic	0.010	0.96666
5	(16,)	relu	0.100	0.96666